

WASP-50b

R-0718

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_241015 · created 2026-07-27T23:40:37+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460598.84 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1717 +/- 0.0093
Transit depth (Rp/Rs)^2	2.95 +/- 0.32 [%]
Orbital Inclination (inc)	84.54 +/- 1.16
Airmass coefficient 1 (a1)	1.025 +/- 0.013
Airmass coefficient 2 (a2)	-0.019 +/- 0.033
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	#2 - [320, 168]
Optimal Aperture	3.36
Optimal Annulus	8.32
Transit Duration (day)	0.076 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1717 $\pm$ 0.0093 vs 0.1404 (deviation 3.26 σ)
Duration fit vs published	0.076 d vs 0.0754 d (deviation 0.05 σ)
Mid-time O-C	-3.1 min
Depth SNR	17.9
Residual scatter	1.25 %

Light curve

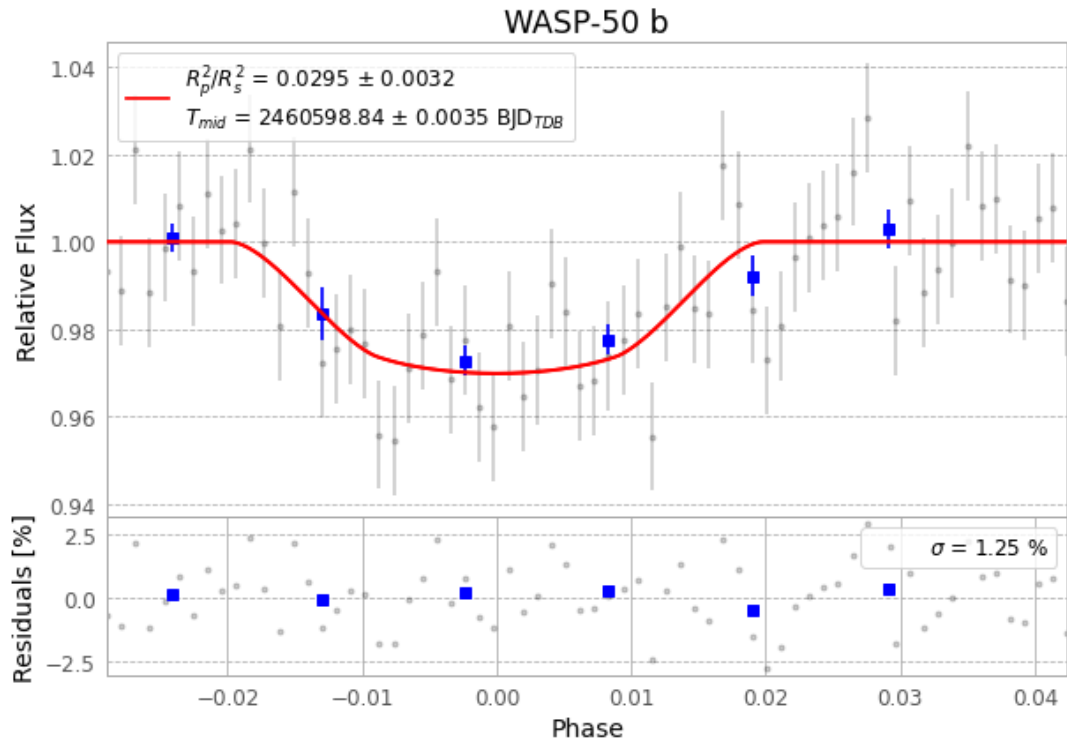


Figure 1 — FinalLightCurve_WASP-50 b_2024-10-14.png (SHA-256 a5c1714371d825f6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [332, 191], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0e74b9451a3cf888...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

68 FITS frames (45 MB), WASP-50241015063915.FITS → WASP-50241015100015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-241015.log (1054d00f4aa5...), FinalLightCurve_WASP-50 b_2024-10-14.png (a5c1714371d8...), FinalLightCurve_WASP-50 b_2024-10-14.csv (7c95d64a04e0...)

Compute resources

Wall time 85.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 23:40Z.